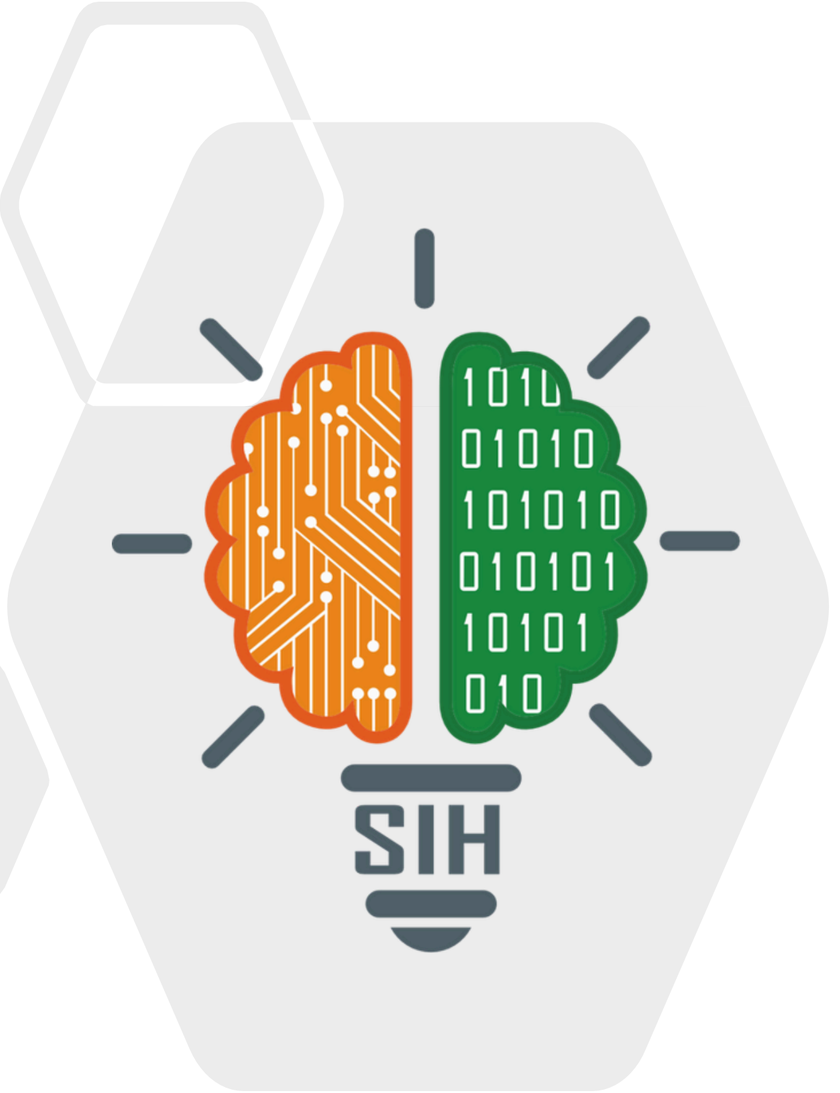
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- **Problem Statement ID –** SIH26233
- **Problem Statement Title-** Inline Microbial Contamination Detection Using Hyperspectral Edge Sensors.
- **Theme -** Agriculture, FoodTech & Rural Development
- **PS Category -** Hardware
- **Team ID -** 141223
- **Team Name -** Spectron Sentinels



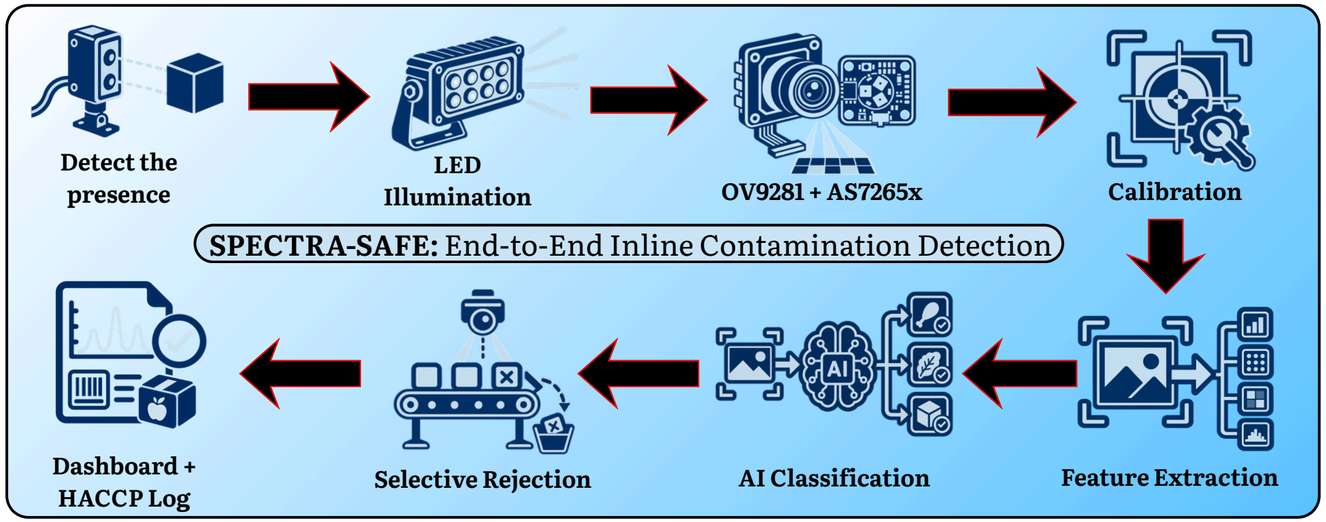
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PROPOSED SOLUTION

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- Wavelength-controlled **275/365/405/510/560/850** nm sequential illumination with an **OV9281** global-shutter camera captures spatial reflectance and fluorescence signatures at 10-bit, up to **120 fps.**
- **AS7265x**-based white/dark drift referencing, flat-field correction and spectral feature extraction feed **PLS-DA** + **Mahalanobis** OOD, while spatial patterns are classified using **YOLO11s-cls** for leafy greens.
- A light-shielded **black chamber** suppresses ambient and stray light, confining 365–850 nm **illumination** to the **sensing zone** for improved **fluorescence SBR** and spectral **repeatability**.
- **RP2040** handles hard real-time strobe, trigger, and encoder timing, while a **Jetson Orin Nano** runs inference—separating deterministic control from AI, with **~70–145 ms** end-to-end latency, fully on-premise.

WORKFLOW:



INNOVATION & UNIQUENESS

1.Wavelength-Selective Multispectral Imaging
Controlled LED bands capture both **spatial** and **fluorescence** signatures for reliable contamination detection.

2.Spectral–Spatial AI Fusion
Combines **spectral** features with **spatial contamination** patterns.

3.Light-Shielded Optical Enclosure
Suppresses **ambient/stray** light for stable **illumination** and **high-contrast fluorescence** detection.

4. Fail-Safe OOD Intelligence
Mahalanobis uncertainty rejects unknown samples and prevents unsafe **false-safe classifications**.

5. Low-Cost Scalable Architecture
Tier-1 multispectral sensing enables cost-effective and scalable **HSI upgrades** for future deployment.

Food	Target / Signature	Excitation	Measurement
Poultry	Fecal / ingesta residue (carrier of Salmonella, Campylobacter)	405 nm	PpIX ~632 nm
Poultry	Ingesta Reflectance Contrast	510/560/850 nm	R565/R517, R517/R802
Greens	Fecal residue vs. chlorophyll	405 nm	F666 / F680 ratio
Greens	Bacterial protein (tryptophan)	275-280 nm	340-350 nm
Biofilm	Biofilm (metabolic NADH)	365 nm	NADH ~440–470 nm

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TECHNICAL APPROACH

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High-Speed Conveyor-Based Hyperspectral Sensing with Edge-AI Classification and Position-Synchronized Rejection

System Architecture

Photo-eye Trigger: 1–2 ms

LED Array: 275 / 365 / 405 / 510 / 560 / 850 nm (strobe)

Spectral Imager: OV9281 (280 x 800 @ 120 fps), 10-bit, global shutter

AS7265x : 410–940 nm, 20 nm FWHM 166 ms @ 16x gain

Reject Actuator: Solenoid + Pneumatic (20–40 ms) into Reject Bin



AI-Powered Multispectral Food Quality Inspection

Hardware Interfacing

LED Illumination Module: 275 / 365 / 405 / 510 / 560 / 850 nm

AS7265x (18-channel): 410–940 nm, 20 nm FWHM 166 ms @ 16x gain

I2C

Rotary Encoder ≥ 1000 PPR

Quad A

Quad B

Photo Eye Sensor : detect the presence of food

Digital

OV9281 Camera: 1280 X 800 @ 120 fps, 10-bit

MIPI CSI-2 (4-lane, ~1.5 Gbps)

MIPI

Optics & Filters:
- 500 nm Long-pass
- 0.75 mm Aperture
- ≥ 20.5° FOV

RP2040 Real-time strobe, encoder and actuator control

CAN

GPIO

Jetson Orin Nano Super (8GB, 67 TOPS)

AI Processing:
- Calibration
- Feature Extraction
- PLS-DA / YOLO11s
- Fusion & Decision

Ethernet

Ethernet-Based Live Dashboard

Storage & Logs: SSD, eMMC, Results + Images, Batch, Traceability

MODBUS TCP

PLC / HMI

Solenoid + Pneumatic Cylinder — Reject Actuator

Target Commodities

Poultry: Fecal residue F632/F600 (78%) R565/R517 (92.5%)

Leafy Greens: Fluorescence F666/F680 (r = 0.96 spinach)

Contact Surfaces: Biofilm 365 nm to NADH to ~ 480 nm (Jun et al. 2010)

Technology stacks

- OpenCV
- PyTorch
- ONNX Runtime
- PLS-DA
- YOLO V11-CLS

Software Pipeline (Jetson Orin Nano Super)

Image Acquisition:

6 frames/item (275 / 365 / 405 / 510 / 560 / 850 nm)

I(x,y) 10-bit → ~7.7 MB /burst via MIPI CSI-2

Radiometric Correction :

$$R(x,y,\lambda) = \frac{I_{raw} - I_{dock}}{I_{white} - I_{dock} + \epsilon}$$

Normalizes Light Variations

Using AS7265x drift %

Before / After

Registration + Segmentation:

Registration + Segmentation: Aligns & Isolates

Encoder-driven pixel registration

NIR-based product mask

Feature Extraction:

Poultry (12 features): R510, R560, R850, F600, F632, F700 4 ratios + 2 spatial stats

Greens (9 features): F666, F680, F705, R850 3 ratios , moisture index , mask fill fraction

Classification:

PLS-DA (poultry): <1 ms compute time

YOLO11s-cls (greens): 11 ms, 95.4% (spinach)

Mahalanobis OOD: 99th percentile

Verdict:

SAFE

SUSPECT

Contaminated

Unknown

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FEASIBILITY AND VIABILITY

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Technical Feasibility:

- **405 nm** excitation → **632 nm PpIX** emission - established **fecal fluorescence** channel.
- **OV9281** global shutter, **1280×800 @ 120 fps** - six-band strobe burst.
- **AS7265x** 18-channel **410–940 nm** reference - corrects **LED drift inline**.

Technical Viability:

- **565/517** reflectance ratio - **92.5% fecal discrimination** reported on poultry.
- **YOLO11s-cls, 11 MB, 11 ms** inference - 92% spinach classification accuracy.
- **Mahalanobis OOD** gate → **UNKNOWN** - defined **fail-safe** rejection target.

Operational Feasibility:

- Non-contact **optical tunnel** - no cutting, stopping, or contact.
- Encoder-tracked rejection - **tolerates** belt speed variation without fixed timers.
- **Light-tight interlocked enclosure** - removes ambient light and UV-C exposure.

FEASIBILITY & VIABILITY

Operational Viability:

- **343–429 ms** per bird at 140–175 **BPM** - **~135 ms** pipeline.
- **Per-SKU** models (poultry, spinach) commodity-specific thresholds, not one.
- **JSON audit record** per item - supports **HACCP** and **FSSAI traceability**.

Economic Feasibility:

- **~₹70,000** Tier-1 build - LEDs, filters, camera, **Jetson**, actuator.
- Multispectral LEDs replace **hyperspectral cameras**—roughly 95% **capital** reduction.
- **Edge inference** only - no cloud, bandwidth, or licence cost.

Economic Viability:

- **Sensor-agnostic** software - **Tier-2** hyperspectral upgrade without rewriting the **pipeline**.
- **100% screening** versus fractional sampling - contamination caught.
- **False reject** costs one **carcass** - one avoided **recall** repays deployment.

Conventional	Our Solution
Random batch sampling	100% inline screening
24–72 h lab result	Verdict in <150 ms
Lab test is sole control	Screening layer plus lab confirmation
Destructive culture	Non-contact, non-destructive optical
Detection after packaging	Rejection before the chiller
Single-band colour vision	6-band UV / VIS / NIR
Hyperspectral camera, ₹15–40 lakh	LED + filter multispectral, ~₹70k
Cloud or offline inference	Jetson Orin Nano edge
One threshold for all produce	Per-commodity calibrated models

VALIDATED TECHNICAL BASIS

1. USDA-FSIS zero visible fecal tolerance on broiler carcasses entering chiller; FSSAI Schedule 4 hygiene norms.
2. Kim et al., J. Food Protection 66(7), 2003 — 410–420 nm excitation, 632 nm PpIX emission optimal.
3. Seo et al., Sensors 19(16):3483, 2019 — on-line multispectral fluorescence imaging at 1–5 birds/second.

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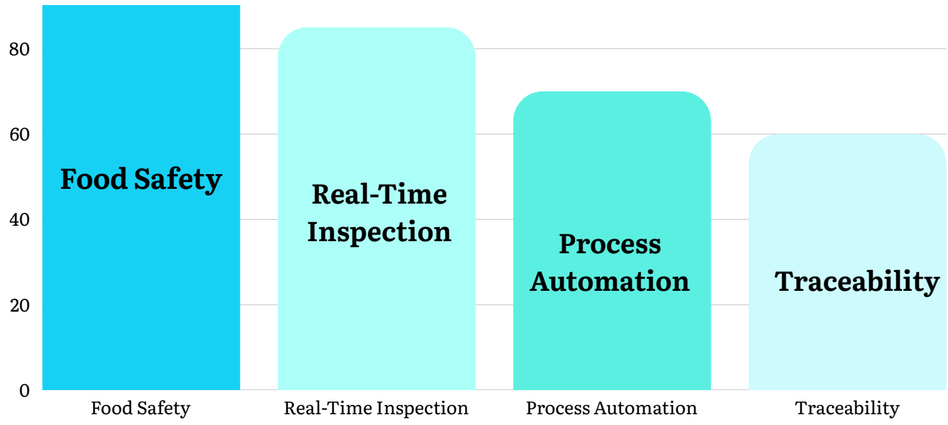
IMPACT AND BENEFITS

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Impact Area	Description	SDG Connection
Enhanced Food Safety	Presumptive screening detects contamination-spectral signatures before product packaging	**SDG 3** – Good Health & Well-being **SDG 12** – Responsible Consumption & Production
Real-Time Inspection	Edge inference enables continuous screening without halting conveyor operation	**SDG 9** – Industry, Innovation & Infrastructure **SDG 12** – Responsible Consumption & Production
Non-Destructive Detection	Spectral inspection identifies contamination without damaging or destroying food products	**SDG 9** – Industry, Innovation & Infrastructure **SDG 12** – Responsible Consumption & Production
Process Automation	Encoder-synchronized rejection automatically isolates flagged items downstream	**SDG 8** – Decent Work & Economic Growth **SDG 9** – Industry, Innovation & Infrastructure
Traceable Compliance	Features, verdict, timestamp and batch ID create auditable safety records	**SDG 12** – Responsible Consumption & Production **SDG 16** – Peace, Justice & Strong Institutions

Metric	Target
Inspection Coverage	100% Items*
Decision Latency	≤ 300 ms*
Throughput	3 items/second*
Decision Priority	Sensitivity-Biased
Decision Traceability	100% Logged*

Key benefits of Spectra-Safe



RESEARCH AND REFERENCES

Purpose: Establishes our violet excitation wavelength selection.
Source Content: 410–420 nm excitation, 632 nm PpIX emission.
Justification: Defines our primary poultry fluorescence channel

Line-scan hyperspectral imaging system for real-time inspection of poultry

Hyperspectral Imaging for Detecting Fecal and Ingesta Contamination on Poultry Carcasses

Purpose: Validates our reflectance band-ratio approach.
Source Content: 565/517 ratio achieved 92.5% discrimination.
Justification: Grounds our multispectral reflectance feature set.

SPECTRA-SAFE

Purpose: Demonstrates conveyor-speed multispectral fluorescence inspection.
Source Content: Line-scan imaging at 1–5 birds per second.
Justification: Proves our inline architecture at processing speed.

Multispectral Fluorescence Imaging for On-Line Inspection of Fecal Residues

Detecting E. coli on Plant Leaf Surfaces Using UV-C Fluorescence and Deep Learning

Purpose: Validates our leafy-green deep learning path.
Source Content: YOLO11s-cls reached 92% accuracy, 11 ms.
Justification: Justifies our spatial classifier and edge deployment.